Spurious Correlation Identification and Debiased Data to Improve Natural Language Inference

Abstract

The goal of this project is twofold. The first goal is to identify spurious correlations, also known as dataset artifacts, between word tokens and class labels in the SNLI dataset. Additionally, we will explore if these biases in the training data manifest themselves in our NLI model predictions. This will closely follow the statistical testing methods explored by Gardner et al., 2021. The second goal is to improve upon our NLI task on the SNLI and SNLI-hard datasets by training a model on a debiased version of the SNLI dataset, namely the SNLI Par-Z debiased dataset produced by Wu et al., 2022. We show that finetuning our ELECTRA-small model on the SNLI Par-Z dataset achieves an accuracy of 80.59% on the SNLI-hard test set, improving on the 77.23% accuracy achieved by an ELECTRA-small model fine-tuned on the base SNLI dataset.

1 Introduction

NLI datasets often contain spurious correlations between single words and labels. For example, samples containing the word “there” may be more frequently labeled “entailment” in a dataset. In other words, the presence of the word “there” in either the premise and/or hypothesis is correlated with likelihood of being classified as entailment. Models can potentially pick up on these correlations to achieve seemingly good performance in sample, only to fail to generalize well to more challenging datasets where these biases may not be present. Utilizing these spurious correlations serves as a sort of undesired short cut for models, as ultimately the model is not achieving true linguistic competence.

One way to identify such dataset artifacts is provided by Gardner et al., 2021 using their “competency problems” framework. This framework leverages the simple, but strong assumption that all simple correlations between input features and output labels are spurious. Under this assumption, the probability of an output label given an input feature must be uniform over all possible output labels. For example, if the word “there” is present in the example the probability for each class (entailment, neutral, contradiction) should be 1/3. The rationale for this assumption is that complex language cannot be boiled down to simple feature correlations. Therefore, we can use the competency problem assumption to test for statistically significant deviations away from the uniform output label distribution.

Armed with z-statistics for each word (i.e. input feature) and class label, we can measure how deviations away from the uniform label distribution manifest themselves in model predictions. Specifically, we can measure the average model predicted class probability for single tokens and compare them between high and low z-score words.

Lastly, with knowledge of the z-statistics for each token and class label we look to mitigate bias in the training data and thus model predictions. The end goal is a model that performs better on a challenging subset of the SNLI data, namely the SNLI-hard test set constructed by Gururangan et al., 2018. The process for constructing a debiased training set is out of the scope of this project. Here we simply utilize the SNLI Par-Z dataset constructed by Wu et al., 2022. It is a mixture of z-filtered SNLI data and samples from a trained data generator, which are in the distribution of the existing SNLI dataset.

2 Looking for dataset artifacts

In this section we analyze the SNLI and SNLI Par-Z training datasets for correlations between word tokens and output labels. We then perform the hypothesis test outlined in Gardner et al., 2021. to identify statistically significant deviations from the uniform conditional class probability distribution.

2.1 Approach

Under the competency problem framework any correlations between single tokens and class labels are assumed to be spurious. To identify these correlations we conduct a hypothesis test on each token in the vocabulary using the uniform conditional class probability. Specifically, we will use the observed proportion $\hat{p}(y|x_i)$ of a class label y, given a word token x_i. The null hypothesis is that the conditional class probability is $\hat{p}(y|x_i) = 1/3$ for all 3 possible labels y: entailment, neutral, and contradiction. The alternative hypothesis is that $\hat{p}(y|x_i) > 1/3$. For each word token the z-statistic will be the following:

$$z^* = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

where $\hat{p}$ is shorthand for the observed proportion, $p_0 = 1/3$, and n is the total number examples in the dataset containing that word token.

To conduct the hypothesis test we construct a vocabulary of all the words present in both premise and hypothesis across all examples. We utilize NTLK work tokenization and keep only tokens consisting entirely of alphabetic characters. Next, we count the number of examples of each word token and class label. This is effectively a contingency table (or crosstab) with the word token along one axis and the class label along the other axis. We then compute the z-statistic and p-value for each word token-label pair. We use the largest class conditional z-score for each word as our test statistic. As done in Gardner et al., 2021, we use a significance level α = 0.1 with a Bonferroni correction for the size of the vocabulary.

2.2 Results

We perform the procedure described in the previous section to both the SNLI train set and SNLI Par-Z set. The results are summarized in Table 1 below. An artifact in this case is a token-label pair whose observed proportion in the dataset is statistically higher than 1/3. In other words, artifacts are words tokens whose maximum class conditional z-score is higher than our critical z value (i.e. reject the null hypothesis). As expected, the SNLI Par-Z contains proportionately fewer artifacts (spurious correlations) than the base SNLI training set.

Dataset	**Size**	**Vocab. size**	**Artifacts**
SNLI	550,152	32,691	3.65%
SNLI Par-Z	933,085	67,011	**1.98%**

Table 1: Training datasets comparison

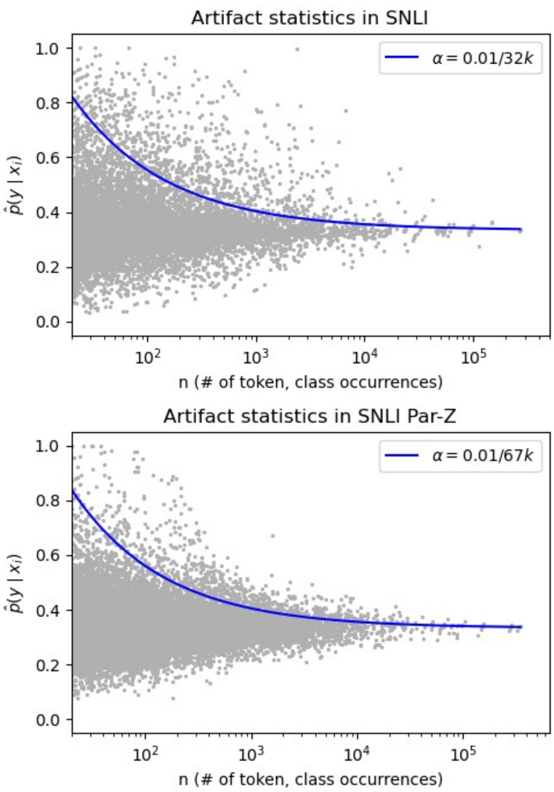

Figure 1: Observed class label proportions in SNLI training set and SNLI Par-Z. Statistically significant deviations from the competency problem are above the blue line.

Figure 1 above shows the distribution of observed class proportions in each training dataset. Each point represents a word-label pair. The observed class proportions $\hat{p}(y|x_i)$ is along the y-axis, and the number of examples, n, is along the x-axis. The blue line represents the Bonferroni-corrected significance threshold of our hypothesis test. Points above the blue line indicate data artifacts. Notably, observed proportions in the SNLI dataset (Fig. 1, top) show a higher variance than the SNLI Par-Z dataset (Fig. 1, bottom). Points are more tightly distributed around $\hat{p}(y|x_i) = 1/3$ in the SNLI Par-Z dataset, with more mass below the blue line. Again, this is expected as the SNLI Par-Z dataset was constructed with this very intention. This was promising as we looked to reduce bias in our model predictions and improve generalizability to more difficult settings.

3 Improving the model with debiased data

Here we turn our attention to the output of two ELECTRA-small models: one fine-tuned on the SNLI training set and another fine-tuned on the SNLI Par-Z dataset, the latter being our debiased dataset. We evaluated each model on three different datasets: a synthetic single token dataset, the SNLI validation set, and the SNLI-hard test set from Gururangan et al., 2018.

The single token dataset serves as way to see if a model will bias its output without the full context (Gardner et al., 2021). It allows us to measure the average model predicted class probability for single tokens and see if there is a difference between highest and lowest z-score tokens. A non-zero difference would indicate our model has learned some of the bias of our datasets.

The experiment on the SNLI-hard dataset is similar to those conducted in Wu et al., 2022 except here we use an ELECTRA-small model, whereas Wu et al. used a BERT-base model.

3.1 Training the models

To perform our model analysis we first fine-tuned two ELECTRA-small models on the aforementioned datasets. The SNLI Par-Z dataset contains about twice as many training examples as SNLI (see Table 1). To keep the models comparable, we limited the maximum training examples on the SNLI Par-Z fine-tuned model to 550,152. This exactly matches the training set size of base SNLI training set. Specifically, the subset of SNLI Par-Z we used in training contains 124,602 z-filtered examples from the original SNLI dataset. The remaining 425,523 examples were generated from the trained data generator in Wu et al., 2022. Again, no z-filtering or data generation was done in this project. We simply utilized a subset of the existing SNLI Par-Z dataset from Wu et al., 2022.

3.2 Measuring learned model bias

To get a sense if our models learned any of the bias present in the datasets we evaluated each of our two models on single token datasets. Here we again followed the model analysis performed in Gardner et al., 2021. For each of our training datasets, SNLI and SNLI Par-Z, we performed the following.

First, we selected the words appearing at least 20 times in the dataset and partitioned these words into 3 groups based on their maximum class conditional z-score. Within each of these 3 groups we select the 50 largest and 50 smallest class conditional z-score words. This left us with 300 words per dataset, partitioned into 3 groups and 2 subgroups within each group. For each of these 300 words we constructed two synthetic examples: one example with only the single word as the premise and an empty hypothesis and another example with an empty premise and single-word hypothesis. In total we have 600 synthetic examples per dataset.

Fine-tuning dataset	$\Delta\hat{p}_y$		
	Entailment	**Neutral**	**Contradiction**
SNLI	36.8%	23.1%	31.3%
SNLI Par-Z	**33.2%**	**7.4%**	**12.9%**

Table 2: Difference in average model predicted class probability between high and low z-score tokens using ELECTRA-small fine-tuned on SNLI and SNLI Par-Z datasets.

Next, we used each of our fine-tuned models to predict the labels on their respective single-token datasets. We then computed the predicted class probabilities by applying a softmax over the logits outputted by the model. Then, for each word we averaged the two predicted class probabilities, since for each word there are two examples: single-word premise with empty hypothesis and empty premise with single-word hypothesis. For each class group partition we computed the average of the predicted probability for the 50 largest z-score words and the average for the 50 smallest z-score words. We then took the difference between these two averages which we denote $\Delta\hat{p}_y$ in Table 2 above. This is our measure for model bias, with a non-zero quantity indicating the presence of a bias.

From Table 2 we see that the model fined-tuned on the SNLI Par-Z dataset exhibits much lower bias on the single token inputs, particularly for the neutral and contradiction classes. In short, there was less bias in the training data and therefore less bias for the model to learn. In the next section we explore how less dataset and model bias affect performance on more challenging datasets, where biased models have previously struggled.

Interestingly, the model bias for the entailment class does not decrease as much as the other two classes. It is not immediately clear why this is the case and is good subject for follow up analysis.

3.3 Evaluating on SNLI-hard

Aside from evaluating model bias on single token inputs, we also tested each model on the SNLI-hard test set from Gururangan et al., 2018. This test set filters the original SNLI dataset for examples which a hypothesis-only model predicted incorrectly.

The model accuracy for each of our two fine-tuned models evaluated on the two datasets is shown below in Table 3. Between the two models the base SNLI fine-tuned model performed better on the SNLI validation set. Yet, the Par-Z fined tuned model performed better than the SNLI fine-tuned model on the SNLI-hard test set.

Fine-tuning dataset	SNLI	SNLI-hard
SNLI	**89.43**	77.23
SNLI Par-Z	87.02	**80.59**

Table 3: Accuracy of models fine-tuned on SNLI and SNLI Par-Z and evaluated on SNLI and SNLI-hard datasets.

Given the SNLI-hard dataset is comprised of examples that a hypothesis-only model got wrong it's possible the SNLI fine-tuned model is also relying on biases in the hypotheses and thus struggles with the SNLI-hard dataset. Once these biases in the hypotheses have been either minimized or removed, such as in the SNLI Par-Z dataset, a model can better learn from the entire context of a training example. This could be an explanation for why the Par-Z fine-tuned model performed better on the SNLI-hard dataset. It was able to learn from both hypothesis and premise, without being inhibited by biases in the hypothesis. This is also supported by the results from section 3.2 and Table 2, where we showed that the Par-Z fine-tuned model exhibited less bias on single token inputs.

4 Conclusion

We showed the presence dataset artifacts in the form of spurious correlations within the SNLI training dataset through hypothesis testing under the competency problem assumption. We demonstrated how these artifacts manifest themselves as model bias by evaluating on single tokens. To mitigate the effect of these artifacts we fine-tuned our ELECTRA-small model on the SNLI Par-Z dataset, which has been constructed to reduce the presence of artifacts. We showed that this achieved an accuracy of 80.59% on the SNLI-hard test. This improves upon the 77.23% accuracy achieved by a model trained on the base SNLI training dataset. This suggests training on debiased datasets can lead to improved performance on challenging settings, where dataset biases may not be present.

References

[Gardner et al., 2021] Matt Gardner, William Merrill, Jesse Dodge, Matthew E Peters, Alexis Ross, Sameer Singh, and Noah Smith. 2021. Competency problems: On finding and removing artifacts in language data. *arXiv preprint arXiv:2104.08646*.

[Wu et al., 2022] Yuxiang Wu, Matt Gardner, Pontus Stenetorp, and Pradeep Dasigi. 2022. Generating Data to Mitigate Spurious Correlations in Natural Language Inference Datasets. *arXiv:2203.12942*

[Gururangan et al., 2018] Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. Annotation Artifacts in Natural Language Inference Data. *arXiv:1803.02324*